Slug ascent at Stromboli

The ascent of a gas slug at Stromboli is modeled using `MagmaFOAM`. Ground deformation signals generated by slug ascent are calculated using `syntheticSignals`, and compared to seismic and strain records in the VLP band (Chouet et al., 2003; Mattia et al., 2021).
Pressure at the boundaries of the fluid system is used to calculate ground displacement at the Earth's surface (Longo et al., 2012).

Simulations

slug1 conduit is 170 m long, and 0.5 m wide. A slug forms and rises to the top of the magma column (movie).

slug2 conduit is 170 m long, 0.5 m wide. At time 0, continuous gas injection starts from the bottom of the conduit (movie).